detailed adaptations to particular circumstances (like the resemblance of leaf insects to leaves), specializations for a particular mode of life (like that of the horses for rapid running and grazing), advances in efficiency of major functions (like flight, or vision, or co-ordination of behaviour), or improvements in general organizational plan (like that of arthropods as against segmented worms, or of placentals as against primitive mammals).

He also deduced the inevitability of divergence or diversification—the fact that any successful type would inevitably diverge into two or more diverse types, each adapted to a somewhat different habitat or niche or way of life. This, too, could have been formulated as a universal biological law, for it operates, as Darwin showed, on every scale, from the formation of geographical races within a single species, through cases like that of the single groundfinch ancestor on the Galapagos splitting into a number of distinct genera and species, to the radiation of a large subclass like that of placental mammals into numerous orders, each with its own way of life, up to the divergence of plants from animals. Indeed, as he also pointed out, diversification is itself a biological advantage, since it enables a given area to support a greater bulk of living matter, and in general makes it possible for life to exploit the resources of the environment more fully. Though later-evolving groups are more highly organized, he rightly argued that we should not expect all groups and types of organisms to evolve in the direction of higher organization. Thus single-celled forms, through the very fact of their small size and rapid reproduction, fill a certain natural niche more successfully than larger multicellular creatures can do.

He further showed that natural selection, by its nature, could never cause or promote the evolution of a character which was primarily of advantage to another species. This is a negative generalization as important in its way as the impossibility of perpetual motion is in physical science.

He anticipated modem evolutionary genetics by deducing that large species (with an abundance of individual members) and large genera containing many species will be more variable than small ones, and more likely to produce new species in the course of evolution.

Of course, his views have often had to be modified in detail. This is especially true when he is discussing heredity and variation, since in his day the mechanisms of genetics and mutation were completely unknown.

There was a period, from about 1895 to about 1925, when Darwinism came in for a great deal of criticism, sometimes violent, from many of the leading biologists of the time. They questioned the very notion that characters such as